		When $\frac{dP}{dR} = 0$, $R = 7$ Power output is maximum $P_{max} = 1.29$ W
21	A	The proton experiences an upward force, $F_E = qE = ma$. Hence, upward acceleration of proton, $a_{proton} = qE / m$. An alpha particle is a helium nucleus that has approximately 4 times the mass and twice the charge of a single proton. Hence, upward acceleration of helium, $a_{He} = 2qE / 4m = 0.5\ a_{proton}$. Using $s_y = u_y t + ½\ a_y t^2$ and $u_y = 0$, time taken by helium to hit the top plate $= \sqrt{2} \times$ time taken by proton to hit the top plate However, since horizontal velocity of helium is half of proton, the helium particle will cover a shorter horizontal distance (70.7% of proton's)